Материалы кафедры ММП факультета ВМК МГУ. Введение в глубокое обучение.

Занятие 14. Эмбеддинги слов, классификация текстов

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Источники:

- Конспект по основам NLP, который подготовили: Феоктистов Дмитрий, Хисматуллин Владимир
- Документация gensim
- **UMAP**: Leland McInnes et al., 2018, примерно 22 тысячи цитирований
- Документация umap
- Выделение смысловых компонент слов с помощью Lasso
- Датасет рецензий с Imdb.
- Документация CatBoost
- Примеры датасетов для классификации текстов
- Документация fasttext
- Документация tokenizers

- Документация datasets
- Для большинства изображений указан источник, откуда оно взято, как правило, являющийся статьей

На занятии мы попробуем на практике то, что изучили на лекции – посмотрим на операции с эмбеддингами, попробуем самостоятельно обучить word2vec и fasttext, а также решить с их помощью прикладную задачу классификации текстов. Кроме того, посмотрим на нейросетевой подход к решению данной задачи, а именно, как использовать MLP и сверточные сети для обработки текстов.

In [1]:
```
!pip install gensim
!pip install catboost
!pip install fasttext
!pip install comet_ml
```

```
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x
86_64.whl.metadata (8.4 kB)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dis
t-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist
-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.1
2/dist-packages (from gensim) (7.5.1)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packag
es (from smart_open>=1.8.1->gensim) (2.1.2)
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_64.whl (27.9 MB)
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0
Installing collected packages: gensim
Successfully installed gensim-4.4.0
Collecting catboost
  Downloading catboost-1.2.10-cp312-cp312-manylinux2014_x86_64.whl.metadata
 (1.2 kB)
Requirement already satisfied: graphviz in /usr/local/lib/python3.12/dist-pac
kages (from catboost) (0.21)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-p
ackages (from catboost) (3.10.0)
Requirement already satisfied: numpy<3.0,>=1.16.0 in /usr/local/lib/python3.1
2/dist-packages (from catboost) (2.0.2)
Requirement already satisfied: pandas<4.0,>=0.24 in /usr/local/lib/python3.1
2/dist-packages (from catboost) (2.2.2)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packag
es (from catboost) (1.16.3)
Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packa
ges (from catboost) (5.24.1)
Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages
 (from catboost) (1.17.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/pytho
n3.12/dist-packages (from pandas<4.0,>=0.24->catboost) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist
-packages (from pandas<4.0,>=0.24->catboost) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/di
st-packages (from pandas<4.0,>=0.24->catboost) (2025.3)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/
dist-packages (from matplotlib->catboost) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist
-packages (from matplotlib->catboost) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.1
2/dist-packages (from matplotlib->catboost) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.1
2/dist-packages (from matplotlib->catboost) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/d
ist-packages (from matplotlib->catboost) (26.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-pa
ckages (from matplotlib->catboost) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/
dist-packages (from matplotlib->catboost) (3.3.2)
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.12/d
ist-packages (from plotly->catboost) (9.1.4)
```

```
Downloading catboost-1.2.10-cp312-cp312-manylinux2014_x86_64.whl (97.1 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 97.1/97.1 MB 6.4 MB/s eta 0:00:00
Installing collected packages: catboost
Successfully installed catboost-1.2.10
Collecting fasttext
  Downloading fasttext-0.9.3.tar.gz (73 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 73.4/73.4 kB 1.7 MB/s eta 0:00:
00
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Collecting pybind11>=2.2 (from fasttext)
  Using cached pybind11-3.0.2-py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: setuptools>=0.7.0 in /usr/local/lib/python3.1
2/dist-packages (from fasttext) (75.2.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packag
es (from fasttext) (2.0.2)
Using cached pybind11-3.0.2-py3-none-any.whl (310 kB)
Building wheels for collected packages: fasttext
  Building wheel for fasttext (pyproject.toml) ... done
  Created wheel for fasttext: filename=fasttext-0.9.3-cp312-cp312-linux_x86_6
4.whl size=4647420 sha256=f1c755673316b67f818595416516e9a6922599602f8088402b4
c79f286c6e2a9
  Stored in directory: /root/.cache/pip/wheels/20/27/95/a7baf1b435f1cbde017ca
bdf1e9688526d2b0e929255a359c6
Successfully built fasttext
Installing collected packages: pybind11, fasttext
Successfully installed fasttext-0.9.3 pybind11-3.0.2
Collecting comet_ml
  Downloading comet_ml-3.57.3-py3-none-any.whl.metadata (4.0 kB)
Collecting dulwich!=0.20.33,<1.0.0,>=0.20.6 (from comet_ml)
  Downloading dulwich-0.25.2-cp312-cp312-manylinux_2_28_x86_64.whl.metadata
 (5.4 kB)
Collecting everett<3.2.0,>=1.0.1 (from everett[ini]<3.2.0,>=1.0.1->comet_ml)
  Downloading everett-3.1.0-py2.py3-none-any.whl.metadata (17 kB)
Requirement already satisfied: jsonschema!=3.1.0,>=2.6.0 in /usr/local/lib/py
thon3.12/dist-packages (from comet_ml) (4.26.0)
Requirement already satisfied: psutil>=5.6.3 in /usr/local/lib/python3.12/dis
t-packages (from comet_ml) (5.9.5)
Collecting python-box<7.0.0 (from comet_ml)
  Downloading python_box-6.1.0-py3-none-any.whl.metadata (7.8 kB)
Requirement already satisfied: requests-toolbelt>=0.8.0 in /usr/local/lib/pyt
hon3.12/dist-packages (from comet_ml) (1.0.0)
Requirement already satisfied: requests>=2.18.4 in /usr/local/lib/python3.12/
dist-packages (from comet_ml) (2.32.4)
Requirement already satisfied: rich>=13.3.2 in /usr/local/lib/python3.12/dist
-packages (from comet_ml) (13.9.4)
Requirement already satisfied: semantic-version>=2.8.0 in /usr/local/lib/pyth
on3.12/dist-packages (from comet_ml) (2.10.0)
Requirement already satisfied: sentry-sdk>=1.1.0 in /usr/local/lib/python3.1
2/dist-packages (from comet_ml) (2.54.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-p
ackages (from comet_ml) (75.2.0)
Requirement already satisfied: simplejson in /usr/local/lib/python3.12/dist-p
ackages (from comet_ml) (3.20.2)
Requirement already satisfied: urllib3>=1.26.8 in /usr/local/lib/python3.12/d
```

```
ist-packages (from comet_ml) (2.5.0)
Requirement already satisfied: wrapt>=1.11.2 in /usr/local/lib/python3.12/dis
t-packages (from comet_ml) (2.1.2)
Requirement already satisfied: wurlitzer>=1.0.2 in /usr/local/lib/python3.12/
dist-packages (from comet_ml) (3.1.1)
Collecting configobj (from everett[ini]<3.2.0,>=1.0.1->comet_ml)
  Downloading configobj-5.0.9-py2.py3-none-any.whl.metadata (3.2 kB)
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dis
t-packages (from jsonschema!=3.1.0,>=2.6.0->comet_ml) (25.4.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/l
ocal/lib/python3.12/dist-packages (from jsonschema!=3.1.0,>=2.6.0->comet_ml)
 (2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.
12/dist-packages (from jsonschema!=3.1.0,>=2.6.0->comet_ml) (0.37.0)
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.12/d
ist-packages (from jsonschema!=3.1.0,>=2.6.0->comet_ml) (0.30.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/pyt
hon3.12/dist-packages (from requests>=2.18.4->comet_ml) (3.4.6)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist
-packages (from requests>=2.18.4->comet_ml) (3.11)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.1
2/dist-packages (from requests>=2.18.4->comet_ml) (2026.2.25)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python
3.12/dist-packages (from rich>=13.3.2->comet_ml) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/pyth
on3.12/dist-packages (from rich>=13.3.2->comet_ml) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-p
ackages (from markdown-it-py>=2.2.0->rich>=13.3.2->comet_ml) (0.1.2)
Requirement already satisfied: typing-extensions>=4.4.0 in /usr/local/lib/pyt
hon3.12/dist-packages (from referencing>=0.28.4->jsonschema!=3.1.0,>=2.6.0->c
omet_ml) (4.15.0)
Downloading comet_ml-3.57.3-py3-none-any.whl (786 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 786.2/786.2 kB 6.8 MB/s eta 0:00:
00
Downloading dulwich-0.25.2-cp312-cp312-manylinux_2_28_x86_64.whl (1.4 MB)
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Downloading everett-3.1.0-py2.py3-none-any.whl (35 kB)
Downloading python_box-6.1.0-py3-none-any.whl (27 kB)
Downloading configobj-5.0.9-py2.py3-none-any.whl (35 kB)
Installing collected packages: everett, python-box, dulwich, configobj, comet
_ml
  Attempting uninstall: python-box
    Found existing installation: python-box 7.4.1
    Uninstalling python-box-7.4.1:
      Successfully uninstalled python-box-7.4.1
Successfully installed comet_ml-3.57.3 configobj-5.0.9 dulwich-0.25.2 everett
-3.1.0 python-box-6.1.0
```

In [2]:

```
import os
import numpy as np
import random

import matplotlib_inline
import matplotlib.pyplot as plt

%matplotlib inline
```

```
np.random.seed(52)
random.seed(52)
```

Модели эмбеддингов на основе предсказаний

Мы будем пользоваться библиотекой gensim, так как в ней удобно реализованы различные методы обучения эмбеддингов: `word2vec`, `glove` и другие.

Кроме того, существуют уже предобученные модели, что позволяет в несколько строк получить качественные эмбеддинги. Посмотрим на список доступных моделей ниже.

In [3]:
```
import gensim
import gensim.downloader
```

In [4]:
```
print(list(gensim.downloader.info()['models'].keys()))
```

```
['fasttext-wiki-news-subwords-300', 'conceptnet-numberbatch-17-06-300', 'word
2vec-ruscorpora-300', 'word2vec-google-news-300', 'glove-wiki-gigaword-50',
'glove-wiki-gigaword-100', 'glove-wiki-gigaword-200', 'glove-wiki-gigaword-30
0', 'glove-twitter-25', 'glove-twitter-50', 'glove-twitter-100', 'glove-twitt
er-200', '__testing_word2vec-matrix-synopsis']
```

Для наглядности мы будем использвовать `word2vec-ruscorpora-300`, так как это обученные эмбеддинги для русского языка.

In [5]:
```
word2vec_rus = gensim.downloader.load('word2vec-ruscorpora-300')
```

```
[==================================================] 100.0% 198.8/198.8MB dow
nloaded
```

In [6]:
```
print(f"Всего слов {len(word2vec_rus.index_to_key)}")
print(f"Размерность {word2vec_rus.vector_size}")

print(f"Примеры слов:", ', '.join(word2vec_rus.index_to_key[:10]))
```

```
Всего слов 184973
Размерность 300
Примеры слов: весь_DET, человек_NOUN, мочь_VERB, год_NOUN, сказать_VERB, врем
я_NOUN, говорить_VERB, становиться_VERB, знать_VERB, самый_DET
```

Можно заметить, что каждое слово имеет суффикс, который отвечает за часть речи (NOUN – существительное, VERB – глагол и тд). Задача определения части речи называется `Part of speach tagging`.

Красивую визуализацию можно найти тут.

Мы же попробуем визуализировать и изучить интересные свойства самостоятельно.

`Визуализация`

Мы знаем, что похожие по смыслу слова должны иметь похожие по метрике эмбеддинги. Для визуализации (понижения размерности) будем пользоваться метод UMAP (Leland McInnes et al., 2018), который реализован в библиотеке umap.

In [7]:
```
import umap
```

Выделим специальные слова, за которыми будем следить. Постараемся выбрать слова из разных категорий.

In [8]:
```
embeddings = word2vec_rus.vectors

spec_key = [
    'москва', 'город', 'берлин', 'страна',    # география
    'яблоко', 'груша', 'апельсин', 'помидор', # овощи и фрукты
    'рабочий', 'строитель', 'водитель',       # специальности
    'щука', 'карась', 'окунь'   # рыбы
]

spec_emb  = np.array(list(word2vec_rus.get_vector(k + "_NOUN") for k in spec
spec_emb.shape
```

Out[8]:
```
(14, 300)
```

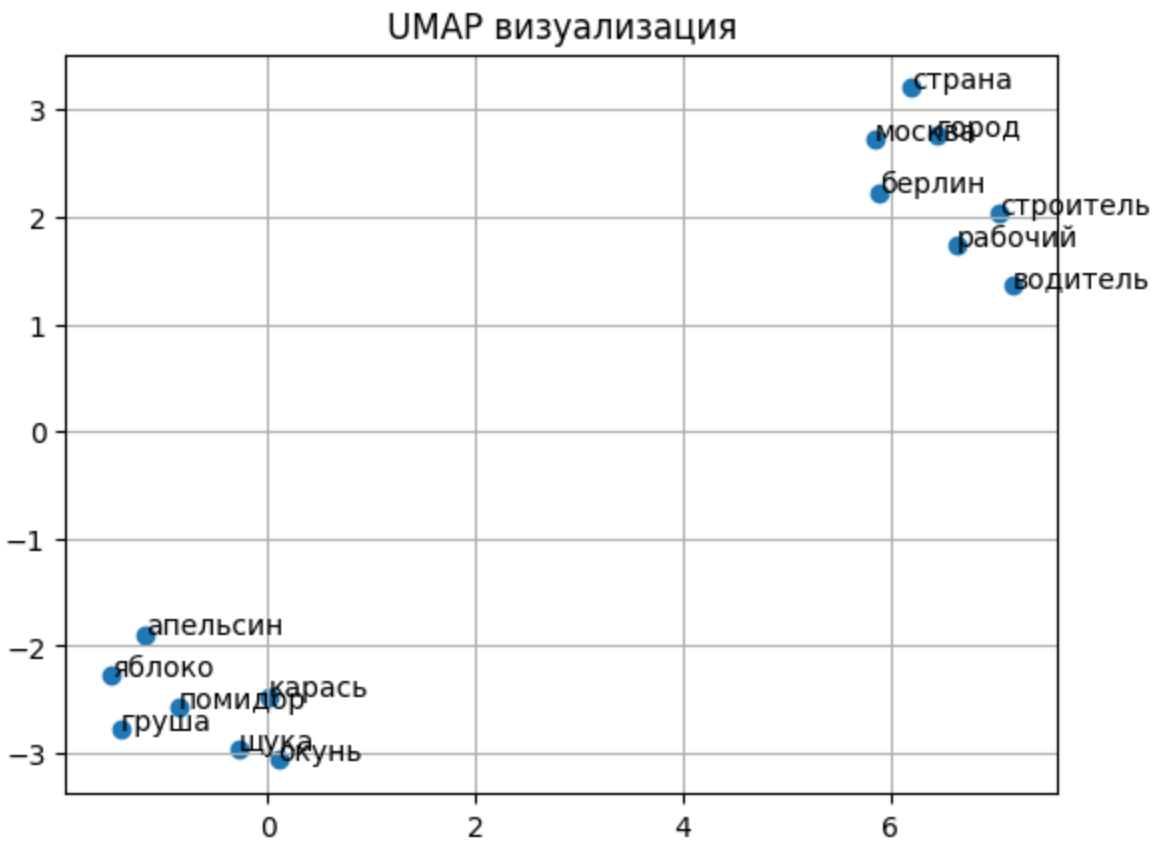
Посмотрим, как расположены выделенные нами слова друг относительно друга.

In [9]:
```
mapper = umap.UMAP(n_components=2, n_neighbors=6, metric='cosine')

u_spec = mapper.fit_transform(spec_emb)
```

In [10]:
```
plt.scatter(u_spec[:,0], u_spec[:,1])

for i, txt in enumerate(spec_key):
    plt.annotate(txt, (u_spec[i, 0], u_spec[i, 1]))
plt.title('UMAP визуализация')
plt.grid()
```



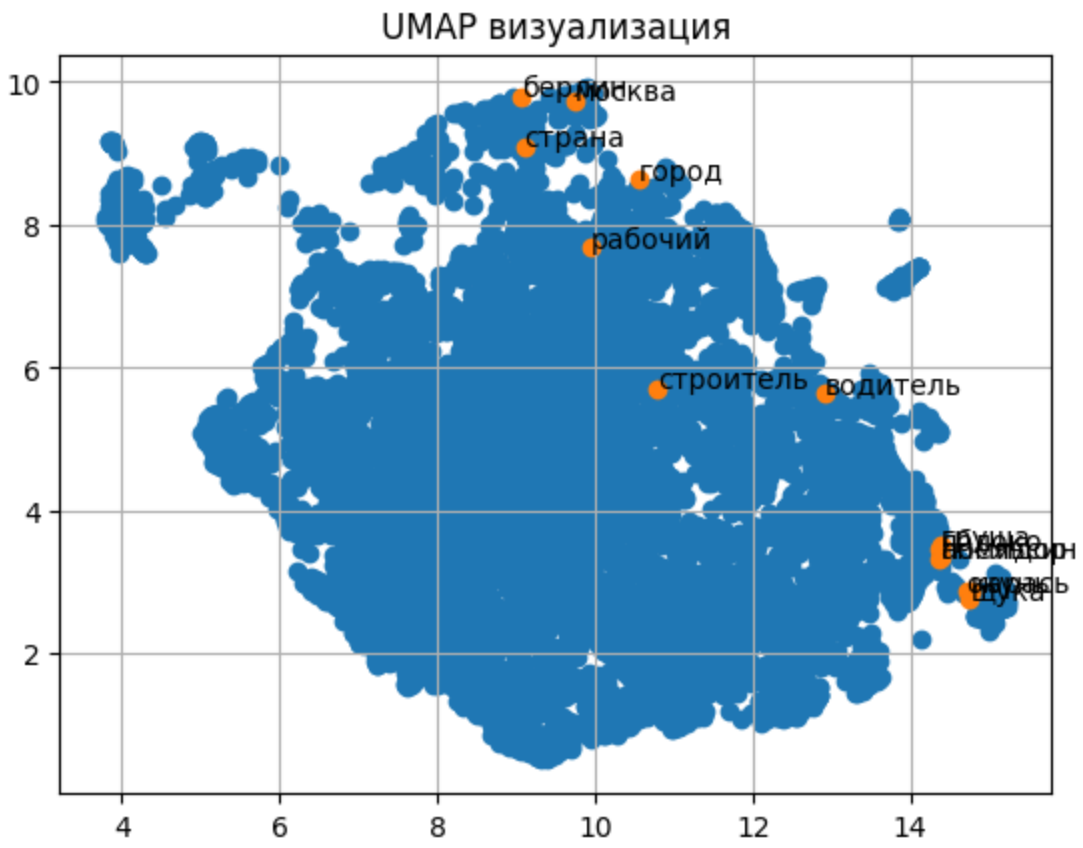
Посмотрим, как расположены выделенные нами слова на общем множестве.

In [11]:
```
mapper = umap.UMAP(n_components=2, metric='cosine')

u      = mapper.fit_transform(embeddings[:10_000])
u_spec = mapper.transform(spec_emb)
```

In [12]:
```
plt.scatter(u[:,0], u[:,1])
plt.scatter(u_spec[:,0], u_spec[:,1])

for i, txt in enumerate(spec_key):
    plt.annotate(txt, (u_spec[i, 0], u_spec[i, 1]))
plt.title('UMAP визуализация')
plt.grid()
```



Операции над эмбеддингами

В моделях из библиотеки `gensim` реализован автоматический поиск наиболее похожих слов, то есть поиск слов, чьи эмбеддинги наиболее близки к эмбеддингу исходного.

In [13]:
```
a = word2vec_rus.get_vector('холод_NOUN')

word2vec_rus.similar_by_vector(a)
```

Out[13]:
```
[('холод_NOUN', 1.0),
 ('стужа_NOUN', 0.754147469997406),
 ('мороз_NOUN', 0.7195620536804199),
 ('сырость_NOUN', 0.6915026307106018),
 ('мерзнуть_VERB', 0.6622565984725952),
 ('тепло_NOUN', 0.6225268840789795),
 ('зябнуть_VERB', 0.6161022186279297),
 ('жара_NOUN', 0.6121419668197632),
 ('замерзать_VERB', 0.6117622256278992),
 ('знобить_VERB', 0.5948620438575745)]
```

Кроме того, мы можем производить над эмбеддингами арифметические операции.

In [14]:
```
a = word2vec_rus.get_vector('король_NOUN')
b = word2vec_rus.get_vector('мужчина_NOUN')

word2vec_rus.similar_by_vector(a - b)
```

Out[14]:
```
[('король_NOUN', 0.6571975350379944),
 ('королевство_NOUN', 0.45206212997436523),
 ('королевский_ADJ', 0.4399426579475403),
 ('максимилианов_NOUN', 0.421102374792099),
 ('кельский_ADJ', 0.41280296444892883),
 ('каролус_NOUN', 0.4126133322715759),
 ('рагоци_NOUN', 0.40673765540122986),
 ('герцог_NOUN', 0.4043634831905365),
 ('королева_NOUN', 0.40316253900527954),
 ('дурлахский_ADJ', 0.4029974937438965)]
```

In [15]:
```
c = word2vec_rus.get_vector('женщина_NOUN')

word2vec_rus.similar_by_vector(a - b + c)
```

Out[15]:
```
[('король_NOUN', 0.8805387616157532),
 ('королева_NOUN', 0.7313904762268066),
 ('герцог_NOUN', 0.6502388715744019),
 ('принцесса_NOUN', 0.6266285181045532),
 ('герцогиня_NOUN', 0.6240381598472595),
 ('королевство_NOUN', 0.6094207167625427),
 ('зюдерманландский_ADJ', 0.6084389686584473),
 ('дурлахский_ADJ', 0.6081665754318237),
 ('ульрик::элеонора_NOUN', 0.6073107719421387),
 ('максимилианов_NOUN', 0.6057003736495972)]
```

In [16]:
```
a = word2vec_rus.get_vector('москва_NOUN')
b = word2vec_rus.get_vector('россия_NOUN')
c = word2vec_rus.get_vector('германия_NOUN')

word2vec_rus.similar_by_vector(a - b + c)
```

Out[16]:
```
[('берлин_NOUN', 0.7166863679885864),
 ('москва_NOUN', 0.6859816312789917),
 ('германия_NOUN', 0.6379823088645935),
 ('мюнхен_NOUN', 0.6141613721847534),
 ('гамбург_NOUN', 0.5910832285881042),
 ('дрезден_NOUN', 0.5751315951347351),
 ('бабельсберг_NOUN', 0.5680847764015198),
 ('париж_NOUN', 0.5643576979637146),
 ('лейпциг_NOUN', 0.555885910987854),
 ('карлсхорст_NOUN', 0.553295373916626)]
```

Чуть аккуратней этот функционал уже реализован в библиотеке.

In [17]:
```
word2vec_rus.most_similar(
    positive=['москва_NOUN', 'германия_NOUN'],
    negative=['россия_NOUN']
)
```

Out[17]:
```
[('берлин_NOUN', 0.7166863679885864),
 ('мюнхен_NOUN', 0.6141614317893982),
 ('гамбург_NOUN', 0.5910831689834595),
 ('дрезден_NOUN', 0.5751316547393799),
 ('бабельсберг_NOUN', 0.5680848360061646),
 ('париж_NOUN', 0.5643577575683594),
 ('лейпциг_NOUN', 0.555885910987854),
 ('карлсхорст_NOUN', 0.5532953143119812),
 ('билефельд_NOUN', 0.5470366477966309),
 ('дуйсбург_NOUN', 0.5449016690254211)]
```

У некоторых слов много значений, поэтому мы можем найти достаточно разные слова по смыслу.

In [18]:
```
word2vec_rus.similar_by_vector('ручка_NOUN')
```

Out[18]:
```
[('ножка_NOUN', 0.6115206480026245),
 ('рукоятка_NOUN', 0.6041494607925415),
 ('рука_NOUN', 0.562088668346405),
 ('пальчик_NOUN', 0.5414899587631226),
 ('рычажок_NOUN', 0.5311201214790344),
 ('ноготочка_NOUN', 0.5253746509552002),
 ('головка_NOUN', 0.5234273076057434),
 ('дверной_ADJ', 0.5118672251701355),
 ('щечка_NOUN', 0.508470356464386),
 ('поприседать_VERB', 0.5080488324165344)]
```

Кроме того, можно быстро находить синонимы.

In [19]:
```
word2vec_rus.similar_by_vector('красивый_ADJ')
```

Out[19]:
```
[('красиво_ADV', 0.6824069619178772),
 ('некрасивый_ADJ', 0.6662992238998413),
 ('нарядный_ADJ', 0.6605890989303589),
 ('прекрасный_ADJ', 0.6481409668922424),
 ('элегантный_ADJ', 0.6363735795021057),
 ('красавец_NOUN', 0.6171321868896484),
 ('очаровательный_ADJ', 0.6072524189949036),
 ('прелестный_ADJ', 0.606392502784729),
 ('хорошенький_ADJ', 0.6016284823417664),
 ('изящный_ADJ', 0.6009820699691772)]
```

Мы ожидаем, что для разных по смыслу словам расстояние между эмбеддингами будет больше, чем между похожими.

In [20]:
```
word2vec_rus.distance('щука_NOUN', 'тумбочка_NOUN')
```

Out[20]:
```
np.float32(0.8575244)
```

In [21]:
```
word2vec_rus.distance('щука_NOUN', 'окунь_NOUN')
```

Out[21]:
```
np.float32(0.2739737)
```

In [22]:
```
word2vec_rus.distance('щука_NOUN', 'карась_NOUN')
```

Out[22]: np.float32(0.31555998)

In [23]:
```
word2vec_rus.distance('щука_NOUN', 'рыба_NOUN')
```

Out[23]: np.float32(0.28943992)

Однако не стоит забывать, что эмбеддинги не идеальные, поэтому иногда возникают странные ситуации.

In [24]:
```
word2vec_rus.distance('щука_NOUN', 'вода_NOUN')
```

Out[24]: np.float32(0.64794594)

В случае более хороших эмбеддингов, полученных из нейронных сетей, можно даже обучать линейные модели для поиска "смысловых компонент" слова. Подробнее можно изучить по ссылке.

Классификация

Мы хотим использовать векторные представления слов для алгоритмов машинного обучения или нейронных сетей. Прежде чем перейти к нейронным сетям, нам нужны бейзлайны – то есть то, с чем будем сравниваться. Сначала проверим, как работают стандартные алгоритмы машинного обучения на эмбеддингах.

Мы будем решать задачу бинарной классификации для датасета рецензий с Imdb.

В качестве базовой модели будем использовать CatBoost.

Подробный список классических датасетов для классификации текста можно посмотреть здесь.

In [25]:
```
if not os.path.exists("aclImdb_v1.tar.gz"):
    ! wget https://ai.stanford.edu/~amaas/data/sentiment/aclImdb_v1.tar.gz
    ! tar -xvf aclImdb_v1.tar.gz
```

Выходные данные были обрезаны до нескольких последних строк (5000).

```
aclImdb/train/unsup/44983_0.txt
aclImdb/train/unsup/44982_0.txt
aclImdb/train/unsup/44981_0.txt
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```

Мы не будем усердно заниматься предобработкой, так как наша цель – именно сравнить методы, а не добиться более высокого качества. В качестве предобработки мы приведём все символы к нижнему регистру.

In [26]:
```
def preprocess(l):
    return l.lower()
```

In [27]:
```
def get_dataset(split='train', base_dir='aclImdb'):
    path_data   = os.path.join(base_dir, split)
    path_data_n = os.path.join(path_data, 'neg')
    path_data_p = os.path.join(path_data, 'pos')
    path_neg  = list(os.listdir(path_data_n))
    path_neg  = sorted(path_neg)
    path_pos  = list(os.listdir(path_data_p))
    path_pos  = sorted(path_pos)


    print(f"Объектов отрицательного класса {len(path_neg)}")
    print(f"Объектов положительного класса {len(path_pos)}")

    X = []
    y = [0] * len(path_neg) + [1] * len(path_pos)

    for p in path_neg:
        with open(os.path.join(path_data_n, p), 'r') as f:
            l = f.readline()
            X.append(preprocess(l))

    for p in path_pos:
        with open(os.path.join(path_data_p, p), 'r') as f:
            l = f.readline()
            X.append(preprocess(l))

    return X, y
```

In [28]:
```
X_train, y_train = get_dataset(split='train')
```

```
Объектов отрицательного класса 12500
Объектов положительного класса 12500
```

In [29]:
```
X_test, y_test = get_dataset(split='test')
```

```
Объектов отрицательного класса 12500
Объектов положительного класса 12500
```

Посмотрим на примеры предложений.

In [30]:
```
X_train[0]
```

Out[30]:
```
"story of a man who has unnatural feelings for a pig. starts out with a ope
ning scene that is a terrific example of absurd comedy. a formal orchestra
audience is turned into an insane, violent mob by the crazy chantings of i
t's singers. unfortunately it stays absurd the whole time with no general n
arrative eventually making it just too off putting. even those from the era
should be turned off. the cryptic dialogue would make shakespeare seem easy
to a third grader. on a technical level it's better than you might think wi
th some good cinematography by future great vilmos zsigmond. future stars s
ally kirkland and frederic forrest can be seen briefly."
```

In [31]:
```
X_test[0]
```

Out[31]:
```
"once again mr. costner has dragged out a movie for far longer than necessa
ry. aside from the terrific sea rescue sequences, of which there are very f
ew i just did not care about any of the characters. most of us have ghosts
in the closet, and costner's character are realized early on, and then forg
otten until much later, by which time i did not care. the character we shou
ld really care about is a very cocky, overconfident ashton kutcher. the pro
blem is he comes off as kid who thinks he's better than anyone else around
him and shows no signs of a cluttered closet. his only obstacle appears to
be winning over costner. finally when we are well past the half way point o
f this stinker, costner tells us all about kutcher's ghosts. we are told wh
y kutcher is driven to be the best with no prior inkling or foreshadowing.
no magic here, it was all i could do to keep from turning it off an hour i
n."
```

Для обучения моделей эмбедингов, нам нужно разделить слова на структурные единицы (символы, слова, N-Gramm). Мы будем работать на уровне слов.

In [32]:
```
def get_corpus(X_data):

    return list(x.split() for x in X_data)
```

In [33]:
```
corpus_train = get_corpus(X_train)
corpus_test  = get_corpus(X_test)
```

In [34]:
```
corpus_train[0][:10]
```

Out[34]:
```
['story', 'of', 'a', 'man', 'who', 'has', 'unnatural', 'feelings', 'for',
'a']
```

word2vec

Обучим word2vec модель на наших данных.

Важно: Мы обучаем модель на всем корпусе текста, так как нам нужно построить словарь. Кроме того, мы решаем unsupervised задачу.

In [35]:
```
main_corpus = corpus_train + corpus_test
```

In [36]:
```
%%time
model_wv = gensim.models.Word2Vec(sentences=main_corpus, vector_size=100, wi
```

```
CPU times: user 4min 2s, sys: 2.2 s, total: 4min 4s
Wall time: 2min 30s
```

Эмбеддинг слова является вектором размерности 100, при этом каждое предложение имеет разную длину. Мы (пока) не умеем обучать модель на данных произвольного размера, поэтому нам нужно как-то аггрегировать эмбединги по всему предложению.

Мы будем использовать средний эмбеддинг предложения, как фичу для обучению классификатора. В таком подходе мы **не учитываем порядок слов**, то есть теряем много информации.

In [37]:
```
def encode_corpus(corpus, encode_fn):

    corp_enc = []
    for sentences in corpus:
        sent_enc = list(encode_fn(w) for w in sentences)
        sent_enc = np.array(sent_enc).mean(axis=0)

        corp_enc.append(sent_enc)

    return np.array(corp_enc)
```

In [38]:
```
enc_train_wv = encode_corpus(corpus_train, lambda k: model_wv.wv[k])
enc_test_wv  = encode_corpus(corpus_test,  lambda k: model_wv.wv[k])
```

In [39]:
```
from catboost import CatBoostClassifier
```

In [40]:
```
def accuracy(y_pred, y_test):
    acc = np.mean(y_pred == y_test)
    print(f"Точность {acc:0.3f}")
```

In [41]:
```
%%time
clf = CatBoostClassifier()
clf.fit(enc_train_wv, y_train, verbose=False)
```

```
CPU times: user 1min 16s, sys: 2.6 s, total: 1min 19s
Wall time: 47.3 s
```

Out[41]:
```
CatBoostClassifier()
```

In [42]:
```
y_pred = clf.predict(enc_test_wv)
accuracy(y_pred, y_test)
```

```
Точность 0.794
```

fasttext

Для использования библиотеки `fasstext` будем использовать специализированную библиотеку. Для этого нам необходимо записать наш корпус текста в файл.

In [43]:
```
import fasttext
```

In [44]:
```
data = X_train + X_test
data = list(d + "\n" for d in data)
```

In [45]:
```
with open("all_data.txt", "w") as f:
    f.writelines(data)
```

In [46]:
```
# Занимает 5 минут
# Progress: 100.0% words/sec/thread:   34499 lr:  0.000000 avg.loss:  2.0042

%%time
model_fasttext = fasttext.train_unsupervised(
    "all_data.txt",
    model='skipgram',
    dim=100,
    epoch=10,
    wordNgrams=2,
    verbose=1
)
```

```
CPU times: user 55min 9s, sys: 5.61 s, total: 55min 15s
Wall time: 55min 43s
```

In [47]:
```
model_fasttext.save_model("model_fasttext.bin")
# model_fasttext = fasttext.load_model("model_fasttext.bin")
```

In [48]:
```
enc_train_ft = encode_corpus(corpus_train, lambda k: model_fasttext[k])
enc_test_ft  = encode_corpus(corpus_test,  lambda k: model_fasttext[k])
```

In [49]:
```
clf = CatBoostClassifier()
clf.fit(enc_train_ft, y_train, verbose=False)
```

Out[49]:
```
CatBoostClassifier()
```

In [50]:
```
y_pred = clf.predict(enc_test_ft)
accuracy(y_pred, y_test)
```

```
Точность 0.847
```

Теперь у нас есть **два бейзлайна**:

- `word2vec` + `CatBoost` – accuracy 0.79
- `fasttext` + `CatBoost` – accuracy 0.85

Нейронный подход

В этой части мы обучим нейронные сети для классификации текстов. Мы рассмотрим, как токенизировать тексты с помощью библиотеки `tokenizers`, чтобы работа с токенами происходила независимо от архитектуры и обучения. Также мы сравним различные архитектуры для классификации, в частности, MLP и свёрточные.

Важной частью будет сравнение результатов с подходами из классического машинного обучения.

Все логи обучения доступны по ссылке.

In [51]:
```
from tqdm import tqdm
from functools import partial

import wandb

import torch
from torch import nn
from torch.utils.data import Dataset, DataLoader

import tokenizers
from tokenizers import Tokenizer, trainers

os.environ["TOKENIZERS_PARALLELISM"] = "false"
```

In [52]:
```
def set_global_seed(seed: int) -> None:
    """Set global seed for reproducibility.
    :param int seed: Seed to be set
    """
    random.seed(seed)
    np.random.seed(seed)

    torch.manual_seed(seed)
    torch.cuda.manual_seed(seed)
    torch.cuda.manual_seed_all(seed)

def print_params_count(model):

    total_params = sum(p.numel() for p in model.parameters())
    total_params_grad = sum(p.numel() for p in model.parameters() if p.requi

    model_name = model.__class__.__name__
    print(f"Информация о числе параметров модели: {model_name}")
    print(f"Всего параметров: \t\t {total_params}")
    print(f"Всего обучаемых параметров: \t {total_params_grad}")
    print()
```

In [53]:
```
set_global_seed(52)
```

tokenizers

Прежде чем создавать модель, нам необходимо создать словарь и произвести токенизацию предложений. Именно от словаря будет зависеть, насколько много "слов" знает наша модель. Мы воспользуемся библиотекой tokenizers. Библиотека `tokenizers` является частью инфраструктуры `Hugging Face`, с который вы познакомитесь на ближайших парах.

Токенезация происходит через класс `tokenizer`. Для того чтобы получить `tokenizer` его надо сначала **обучить**, для этого нам необходимо использовать `tokenizers.trainers`. Так как мы будем работать на уровне слов, то выберем `tokenizers.trainers.WordLevelTrainer`.

Для работы с текстами нам необходимо зарезервировать два специальных токена:

1. `<pad>` для токена означающего паддинг
2. `<unk>` для токенов, которые отсутствуют в словаре

`WordLevelTrainer`

Для начала мы будем разбивать предложение по словам, для этого воспользуемся `trainers.WordLevelTrainer`. Будем рассматривать словарь размером 5000 слов.

```
trainers.WordLevelTrainer(self, /, *args, **kwargs)
Docstring:
Trainer capable of training a WorldLevel model

Args:
    vocab_size (:obj:`int`, `optional`):
        The size of the final vocabulary, including all tokens and
alphabet.

    min_frequency (:obj:`int`, `optional`):
        The minimum frequency a pair should have in order to be
merged.

    show_progress (:obj:`bool`, `optional`):
        Whether to show progress bars while training.

    special_tokens (:obj:`List[Union[str, AddedToken]]`):
        A list of special tokens the model should know of.
```

In [54]:
```
top_n_words = 5_000
```

In [55]:
```
trainer = trainers.WordLevelTrainer(
    vocab_size     = top_n_words,
    special_tokens = ["<pad>", "<unk>"],
    show_progress  = True
)

tokenizer = tokenizers.Tokenizer(
```

```
    model=tokenizers.models.WordLevel()
)
```

Мы будем обучаться из памяти, поэтому нам необходимо передать итератор, который пройдет по всем текстам, которые уже разбиты на слова.

In [56]:
```
text_data = corpus_train + corpus_test
text_data[0][:10]
```

Out[56]:
```
['story', 'of', 'a', 'man', 'who', 'has', 'unnatural', 'feelings', 'for',
'a']
```

In [57]:
```
tokenizer.train_from_iterator(tqdm(text_data, total=len(text_data)), trainer
```

```
100%|██████████| 50000/50000 [00:25<00:00, 1970.70it/s]
```

Посмотрим на словарь.

In [58]:
```
print(f"Размер словаря {tokenizer.get_vocab_size()}")

print(list(tokenizer.get_vocab().keys())[:10])

for idx in [0, 1, 2, 21, 42]:
    print(f"idx = {idx}, word = {tokenizer.id_to_token(idx)}")
```

```
Размер словаря 5000
['harris', 'korean', 'store', 'said,', 'warn', '/>in', 'enjoy', 'hands', 'hol
ly', 'neil']
idx = 0, word = <pad>
idx = 1, word = <unk>
idx = 2, word = the
idx = 21, word = are
idx = 42, word = about
```

Важно: Специальные токены имеют наименьшие *id* по умолчанию.

Для кодирования предложений используется метод `encode`, так как мы самостоятельно описали функцию `tokenize`, то установим `is_pretokenized=True`

In [59]:
```
tokenized_text = text_data[0][:20]

print("Исходный текст:")
print(tokenized_text)

result = tokenizer.encode(tokenized_text, is_pretokenized=True)
print(result)

print("Tokens ids: ", result.ids)
```

```
Исходный текст:
['story', 'of', 'a', 'man', 'who', 'has', 'unnatural', 'feelings', 'for',
'a', 'pig.', 'starts', 'out', 'with', 'a', 'opening', 'scene', 'that', 'is',
'a']
Encoding(num_tokens=20, attributes=[ids, type_ids, tokens, offsets, attention
_mask, special_tokens_mask, overflowing])
Tokens ids:  [77, 5, 3, 160, 36, 41, 1, 1572, 17, 3, 1, 478, 47, 16, 3, 557,
147, 11, 7, 3]
```

Для декодирования воспользуемся методом `decode`, мы хотим посмотреть насколько словарь из 5 тысяч слов выразительный, то есть какие слова мы пропускаем и понятен ли смысл предложение без них.

In [60]:
```
print("Исходный текст:")
print(tokenized_text)

print("Результат декодирования:")
print(tokenizer.decode(result.ids, skip_special_tokens=False))
```

```
Исходный текст:
['story', 'of', 'a', 'man', 'who', 'has', 'unnatural', 'feelings', 'for',
'a', 'pig.', 'starts', 'out', 'with', 'a', 'opening', 'scene', 'that', 'is',
'a']
Результат декодирования:
story of a man who has <unk> feelings for a <unk> starts out with a opening s
cene that is a
```

В примере выше мы видим, что в нашем словаре нет слова `unnatural` и слова `pig.`. Возможно, если бы мы лучше предобратали наш текст, то таких проблем не возникло бы.

Данные

Датасет

В библиотеке `Hugging Face` существуют различные модули, которые помогают оборачивать токенизаторы в удобные датасеты (мы будем пользоваться Datasets). Такой подход позволяет описать обучение сложных моделей в 5-10 строчек кода. В этом ноутбуке мы не сами опишем работу с данными.

Часто бывает полезно ограничить длину предложений параметром `max_len`, чтобы подавать в модель не слишком длинные тексты.

In [61]:
```
class TextDataset(Dataset):
    def __init__(self, texts, targets, tokenizer, max_len=20):
        super().__init__()

        self.max_len   = max_len
        self.texts     = texts
        self.targets   = targets
        self.tokenizer = tokenizer
        self.tokens    = []
```

```
        for text in tqdm(texts):
            tokens = self.tokenizer.encode(text, is_pretokenized=True).ids
            self.tokens.append(tokens)

    def __getitem__(self, idx):
        """
        :param int idx: index of object in dataset
        :return dict: Dictionary with all useful object data
            {
                'text' str: unprocessed text,
                'label' torch.Tensor(dtype=torch.long): sentiment of the tex
                'rating' torch.Tensor(dtype=torch.long): rating of the text
                'tokens' torch.Tensor(dtype=torch.long): tensor of tokens ic
                'tokens_len' torch.Tensor(dtype=torch.long): number of toker
            }
        """

        tokens = self.tokens[idx]
        tokens = tokens[:self.max_len]

        return {
            'text': self.texts[idx],
            'target': torch.tensor(self.targets[idx], dtype=torch.long),
            'tokens': torch.tensor(tokens, dtype=torch.long),
            'tokens_len': torch.tensor(len(tokens), dtype=torch.long),
        }

    def __len__(self):
        """
        :return int: number of objects in dataset
        """
        return len(self.targets)
```

In [62]:
```
train_dataset = TextDataset(corpus_train, y_train, tokenizer)
```

```
100%|██████████| 25000/25000 [00:21<00:00, 1175.27it/s]
```

Посмотрим на данные.

In [63]:
```
train_dataset[0]
```

```
Out[63]: {'text': ['story',
           'of',
           'a',
           'man',
           'who',
           'has',
           'unnatural',
           'feelings',
           'for',
           'a',
           'pig.',
           'starts',
           'out',
           'with',
           'a',
           'opening',
           'scene',
           'that',
           'is',
           'a',
           'terrific',
           'example',
           'of',
           'absurd',
           'comedy.',
           'a',
           'formal',
           'orchestra',
           'audience',
           'is',
           'turned',
           'into',
           'an',
           'insane,',
           'violent',
           'mob',
           'by',
           'the',
           'crazy',
           'chantings',
           'of',
           "it's",
           'singers.',
           'unfortunately',
           'it',
           'stays',
           'absurd',
           'the',
           'whole',
           'time',
           'with',
           'no',
           'general',
           'narrative',
           'eventually',
           'making',
```

```
'it',
'just',
'too',
'off',
'putting.',
'even',
'those',
'from',
'the',
'era',
'should',
'be',
'turned',
'off.',
'the',
'cryptic',
'dialogue',
'would',
'make',
'shakespeare',
'seem',
'easy',
'to',
'a',
'third',
'grader.',
'on',
'a',
'technical',
'level',
"it's",
'better',
'than',
'you',
'might',
'think',
'with',
'some',
'good',
'cinematography',
'by',
'future',
'great',
'vilmos',
'zsigmond.',
'future',
'stars',
'sally',
'kirkland',
'and',
'frederic',
'forrest',
'can',
'be',
'seen',
'briefly.'],
```

```
 'target': tensor(0),
 'tokens': tensor([  77,    5,    3,  160,   36,   41,    1, 1572,   17,
3,    1,  478,
           47,   16,    3,  557,  147,   11,    7,    3]),
 'tokens_len': tensor(20)}
```

In [64]:
```
train_dataset[-1]
```

```
Out[64]: {'text': ['working-class',
           'romantic',
           'drama',
           'from',
           'director',
           'martin',
           'ritt',
           'is',
           'as',
           'unbelievable',
           'as',
           'they',
           'come,',
           'yet',
           'there',
           'are',
           'moments',
           'of',
           'pleasure',
           'due',
           'mostly',
           'to',
           'the',
           'charisma',
           'of',
           'stars',
           'jane',
           'fonda',
           'and',
           'robert',
           'de',
           'niro',
           '(both',
           'terrific).',
           "she's",
           'a',
           'widow',
           'who',
           "can't",
           'move',
           'on,',
           "he's",
           'illiterate',
           'and',
           'a',
           'closet-inventor--you',
           'can',
           'probably',
           'guess',
           'the',
           'rest.',
           'adaptation',
           'of',
           'pat',
           "barker's",
           'novel',
```

```
'"union',
'street"',
'(a',
'better',
'title!)',
'is',
'so',
'laid-back',
'it',
'verges',
'on',
'bland,',
'and',
'the',
"film's",
'editing',
'is',
'a',
'mess,',
'but',
"it's",
'still',
'pleasant;',
'a',
'rosy-hued',
'blue-collar',
'fantasy.',
'there',
'are',
'no',
'overtures',
'to',
'serious',
'issues',
'(even',
'the',
'illiteracy',
'angle',
'is',
'just',
'a',
'plot-tool',
'for',
'the',
'ensuing',
'love',
'story)',
'and',
'no',
'real',
'fireworks,',
'though',
'the',
'characters',
'are',
'intentionally',
```

```
  'a',
  'bit',
  'colorless',
  'and',
  'the',
  'leads',
  'are',
  'toned',
  'down',
  'to',
  'an',
  'interesting',
  'degree.',
  'the',
  'finale',
  'is',
  'pure',
  'fluff--and',
  'cynics',
  'will',
  'find',
  'it',
  'difficult',
  'to',
  'swallow--though',
  'these',
  'two',
  'characters',
  'deserve',
  'a',
  'happy',
  'ending',
  'and',
  'the',
  'picture',
  "wouldn't",
  'really',
  'be',
  'satisfying',
  'any',
  'other',
  'way.',
  '***',
  'from',
  '****'],
 'target': tensor(1),
 'tokens': tensor([   1,  749,  649,   34,  204, 1682,    1,    7,   15, 20
80,   15,   33,
            1,  265,   51,   21,  472,    5, 2111,  618]),
 'tokens_len': tensor(20)}
```

Даталоадер

Даталоадер предназначен для того, чтобы эффективно подавать данные из датасета в нашу модель. Он разбивает их на батчи и перемешивает их, а также реализует

функционал параллельной подгрузки батчей на device. Чтобы объединить несколько тензоров разной длины в один можно использовать функцию `torch.nn.utils.rnn.pad_sequence`

Обратите внимание на её аргументы:

1. `batch_first` определяет по какой оси "складывать" тензоры. Мы будем использовать `batch_first=True`, так как это удобнее для восприятия.
2. `padding_value` – число, которое будет использоваться в качестве паддинга, чтобы сделать все тензоры одинаковой длины

In [65]:
```
torch.nn.utils.rnn.pad_sequence([
    torch.tensor([1, 2, 3]),
    torch.tensor([4, 5]),
    torch.tensor([6, 7, 8, 9])
], batch_first=True, padding_value=-1)
```

Out[65]:
```
tensor([[ 1,  2,  3, -1],
        [ 4,  5, -1, -1],
        [ 6,  7,  8,  9]])
```

In [66]:
```
def collate_fn(batch, padding_value, batch_first=True):
    """
    :param List[Dict] batch: List of objects from dataset
    :param int padding_value: Value that will be used to pad tokens
    :param bool batch_first: If True resulting tensor with tokens must have
    :return dict: Dictionary with all data collated
        {
            'ratings' torch.Tensor(dtype=torch.long): rating of the text for
            'labels' to rch.Tensor(dtype=torch.long): sentiment of the text

            'texts' List[str]: All texts in one list
            'tokens' torch.Tensor(dtype=torch.long): tensor of tokens ids pa
            'tokens_lens' torch.Tensor(dtype=torch.long): number of tokens f
        }
    """

    texts       = [obj['text'] for obj in batch]
    targets     = torch.stack([obj['target'] for obj in batch])
    tokens_lens = torch.stack([obj['tokens_len'] for obj in batch])
    tokens      = torch.nn.utils.rnn.pad_sequence(
        [obj['tokens'] for obj in batch], batch_first=batch_first, padding_v
    )

    return {
        'targets': targets,

        'texts': texts,
        'tokens': tokens,
        'tokens_lens': tokens_lens
    }
```

In [67]:
```
train_dataloader = DataLoader(
```

```
    train_dataset, batch_size=4, shuffle=True, num_workers=0,
    collate_fn=partial(collate_fn, padding_value=tokenizer.token_to_id('<pac
)
```

Посмотрим на какой-нибудь батч:

In [68]:
```
batch = next(iter(train_dataloader))
batch.keys(), batch['targets'], batch['tokens'], batch['tokens_lens']
```

Out[68]:
```
(dict_keys(['targets', 'texts', 'tokens', 'tokens_lens']),
 tensor([1, 0, 1, 1]),
 tensor([[  10,    7,   30,    5,  135,  123,   11,   24,  337,  566,   42,
   50,
            24, 3915,   63,    2,  364,    1,   12,   74],
         [  61,  610,  142,  252, 1188,   10,    7,    3,   56,   25,  334,
  817,
           229,    7,    3,  307,    1,    9,   37, 1751],
         [  94,  295,  380,  123,    5,    1,    9,   62,    3,  174,  798,
1192,
            42,  102,  315,   27,    1,  105,   75,   19],
         [1522, 2924,    7,    3,  406,  691,    5,    3,    1,  138,   12,
2033,
          1041,  782,   16,  319,  887,    4,    1,  791]]),
 tensor([20, 20, 20, 20]))
```

Эмбеддинги в PyTorch

В случае обучение нейронной сети мы будем сами обучать эмбеддинги токенов. Для этого нам нужно использовать класс `torch.nn.Embedding`, который по сути реализует массив.

Эмбеддинг токена i это строка i в матрице эмбеддингов.

- `num_embeddings` – число токенов
- `embedding_dim` – размерность эмбеддинга
- `padding_idx` – индекс токена `<pad>`

```
torch.nn.Embedding(
    num_embeddings: int,
    embedding_dim: int,
    padding_idx: Optional[int] = None,
    max_norm: Optional[float] = None,
    norm_type: float = 2.0,
    scale_grad_by_freq: bool = False,
    sparse: bool = False,
    _weight: Optional[torch.Tensor] = None,
    _freeze: bool = False,
    device=None,
    dtype=None,
) -> None
```

Посмотрим как преобразуется батч после слоя `nn.Embedding`.

In [69]:
```
embed_dim = 128

emb_layer = torch.nn.Embedding(
    tokenizer.get_vocab_size(), embed_dim,
    padding_idx=tokenizer.token_to_id('<pad>')
)
```

In [70]:
```
out = emb_layer(batch['tokens'])

out.shape
```

Out[70]:
```
torch.Size([4, 20, 128])
```

Так как `nn.Embedding` реализует матрицу, то число параметров зависит от размера словаря и размерности эмеддингов.

In [71]:
```
tokenizer.get_vocab_size() * embed_dim
```

Out[71]:
```
640000
```

In [72]:
```
print_params_count(emb_layer)
```

```
Информация о числе параметров модели: Embedding
Всего параметров:                640000
Всего обучаемых параметров:      640000
```

Полносвязная нейронная сеть (MLP)

В этой части мы обучим классификатор текстов на основе полносвязной нейронной сети. Выше мы уже создали удобные классы-обёртки для работы с данными. Теперь мы соберем модель для решения задачи классификации.

In [73]:
```
class MLPClassifier(torch.nn.Module):
    def __init__(
        self, embedding_dim, hidden_dim, output_size, tokenizer: Tokenizer,
        dropout=0.5
    ):
        super().__init__()

        self.dropout = dropout

        self.tokenizer = tokenizer
        self.hidden_dim = hidden_dim
        self.output_size = output_size
        self.embedding_dim = embedding_dim

        # Create a simple lookup table that stores embeddings of a fixed dic
        #    Use torch.nn.Embedding.
        self.word_embeddings = torch.nn.Embedding(
```

```
            self.tokenizer.get_vocab_size(), self.embedding_dim,
            padding_idx=self.tokenizer.token_to_id('<pad>')
        )

        self.features = torch.nn.Sequential(
            nn.Linear(self.embedding_dim, self.hidden_dim),
            nn.ReLU(), nn.Dropout1d(self.dropout),
            nn.Linear(self.hidden_dim,    self.hidden_dim),
            nn.ReLU(), nn.Dropout1d(self.dropout),
            nn.Linear(self.hidden_dim, self.hidden_dim),
            nn.ReLU(), nn.Dropout1d(self.dropout),
        )
        # Create linear layer for classification
        self.output = torch.nn.Sequential(
            torch.nn.Linear(self.hidden_dim, self.output_size),
        )

    def forward(self, tokens):
        """
        :param torch.Tensor(dtype=torch.long) tokens: Batch of texts represe
        :return torch.Tensor(dtype=torch.long): Vector representation for ea
        """
        # Evaluate embeddings
        embedding = self.word_embeddings(tokens)

        # Make forward pass through MLP network
        h   = embedding.mean(axis=1)
        h   = self.features(h)
        out = self.output(h)

        return out
```

In [74]:
```
net = MLPClassifier(
    embedding_dim = 128,
    hidden_dim    = 64,
    output_size   = 2,
    tokenizer     = tokenizer
)

out = net(batch['tokens'])
out.shape
```

Out[74]:
```
torch.Size([4, 2])
```

In [75]:
```
print_params_count(net)
```

```
Информация о числе параметров модели: MLPClassifier
Всего параметров:                656706
Всего обучаемых параметров:      656706
```

Цикл обучения

In [77]:
```
set_global_seed(52)
```

```
device = 'cpu'

if torch.cuda.is_available():
    device = 'cuda:0'
device
```

Out[77]: 'cpu'

Мы описали датасет, даталоадер, архитектуру, следовательно можем переходить к циклу обучения. Мы решаем задачу классификации, поэтому будем использовать `nn.CrossEntropyLoss`

In [78]:
```
loss_fn = nn.CrossEntropyLoss(reduction='mean')
```

Для обучения мы будем брать `max_len=100`, то есть оценивать рецензию по первым 100 словам.

In [79]:
```
max_len    = 100
batch_size = 128
lr         = 3 * 1e-3

embedding_dim = 128
hidden_dim    = 256
output_size   = 2
```

In [80]:
```
train_dataset = TextDataset(corpus_train, y_train, tokenizer, max_len=max_le
test_dataset  = TextDataset(corpus_test , y_train, tokenizer, max_len=max_le
```

```
100%|██████████| 25000/25000 [00:33<00:00, 756.47it/s]
100%|██████████| 25000/25000 [00:25<00:00, 987.51it/s]
```

In [81]:
```
train_dataloader = DataLoader(
    train_dataset, batch_size=batch_size, shuffle=True, num_workers=0, drop_
    collate_fn=partial(collate_fn, padding_value=tokenizer.token_to_id('<pac
)
```

In [82]:
```
test_dataloader = DataLoader(
    test_dataset, batch_size=batch_size, shuffle=True, num_workers=0, drop_l
    collate_fn=partial(collate_fn, padding_value=tokenizer.token_to_id('<pac
)
```

Модель мы определили выше, для оптимизации будем использовать `Adam`.

In [84]:
```
set_global_seed(52)

net = MLPClassifier(
    embedding_dim = embedding_dim,
    hidden_dim    = hidden_dim,
    output_size   = output_size,
    tokenizer     = tokenizer
)
```

In [85]:
```
optimizer = torch.optim.Adam(net.parameters(), lr=lr)
```

Цикл обучения возьмем с прошлой пары про `train loop`, поменяем только распаковку батча. В целом, хороший цикл обучения должен быть универсальным для разных задач.

In [86]:
```
from google.colab import userdata
import comet_ml

@torch.no_grad()
def evaluate(net, valid_dataloader, loss_fn, device):

    net.eval()
    loss, accuracy = 0, 0
    count = 0
    for batch in valid_dataloader:
        X_batch = batch['tokens'].to(device)
        y_true  = batch['targets'].to(device)

        out = net(X_batch)
        y_pred = torch.argmax(out, 1)

        bs = out.shape[0]
        loss     += loss_fn(out, y_true).item() * bs
        accuracy += torch.sum(y_pred == y_true).item()
        count    += bs

    return loss / count, accuracy / count



def train_comet_eval(epoch_num, net, optimizer, loss_fn, train_dataloader, v

    experiment = comet_ml.start(
        api_key=userdata.get('COMET_API_KEY'),
        project_name="text_classifier_mmp2026",
        online=True
    )
    experiment.set_name(run_name)

    global_step = 0
    net = net.to(device)

    for _ in tqdm(range(epoch_num)):
        net.train()

        for batch in train_dataloader:
            optimizer.zero_grad()

            X_batch = batch['tokens'].to(device)
            y_true  = batch['targets'].to(device)

            out = net(X_batch)

            loss = loss_fn(out, y_true)
            loss.backward()
```

```
            optimizer.step()

            y_pred = torch.argmax(out, 1)
            accuracy = torch.sum(y_pred == y_true) / y_pred.shape[0]

            experiment.log_metric("train/loss", loss.item(), step=global_ste
            experiment.log_metric("train/accuracy", accuracy.item(), step=gl

            global_step += 1

        loss, accuracy = evaluate(
            net              = net,
            valid_dataloader = valid_dataloader,
            loss_fn          = loss_fn,
            device           = device
        )

        experiment.log_metric("eval/loss", loss, step=global_step)
        experiment.log_metric("eval/accuracy", accuracy, step=global_step)

    experiment.end()
```

In [87]:

```
train_comet_eval(
    epoch_num        = 100,
    net              = net,
    optimizer        = optimizer,
    loss_fn          = loss_fn,
    train_dataloader = train_dataloader,
    valid_dataloader = test_dataloader,
    device           = device,
    run_name         = "MLP_classifier"
)
```

```
COMET WARNING: To get all data logged automatically, import comet_ml before t
he following modules: keras, sklearn, torch, tensorflow, xgboost.
COMET WARNING: As you are running in a Jupyter environment, you will need to
 call `experiment.end()` when finished to ensure all metrics and code are log
ged before exiting.
COMET INFO: Experiment is live on comet.com https://www.comet.com/eadenisoff/
text-classifier-mmp2026/4109615dc1794592af549feecb08ff36

  0%|          | 0/100 [00:00<?, ?it/s]COMET INFO: Couldn't find a Git reposi
tory in '/content' nor in any parent directory. Set `COMET_GIT_DIRECTORY` if
 your Git Repository is elsewhere.
100%|██████████| 100/100 [14:12<00:00,  8.53s/it]
COMET INFO: ------------------------------------------------------------------
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COMET INFO: Comet.ml Experiment Summary
COMET INFO: ------------------------------------------------------------------
----------------------
COMET INFO:   Data:
COMET INFO:     display_summary_level : 1
COMET INFO:     name                  : MLP_classifier
COMET INFO:     url                   : https://www.comet.com/eadenisoff/text
-classifier-mmp2026/4109615dc1794592af549feecb08ff36
COMET INFO:   Metrics [count] (min, max):
COMET INFO:     eval/accuracy [100]    : (0.6376602564102564, 0.7992788461538
461)
COMET INFO:     eval/loss [100]        : (0.4629632003796406, 0.6790351812656
109)
COMET INFO:     train/accuracy [19500] : (0.3671875, 0.734375)
COMET INFO:     train/loss [19500]     : (0.5295639634132385, 0.7985643744468
689)
COMET INFO:   Others:
COMET INFO:     Name         : MLP_classifier
COMET INFO:     notebook_url : https://colab.research.google.com/notebook#fil
eId=1PLmUie0LxPsu0cv-QAVB0Smqqls4aL5w
COMET INFO:   Uploads:
COMET INFO:     environment details : 1
COMET INFO:     filename            : 1
COMET INFO:     installed packages  : 1
COMET INFO:     notebook            : 2
COMET INFO:     os packages         : 1
COMET INFO:     source_code         : 1
COMET INFO:
COMET WARNING: To get all data logged automatically, import comet_ml before t
he following modules: keras, sklearn, torch, tensorflow, xgboost.
```

Теперь мы можем пополнить нашу таблицу результатов:

- `word2vec` + `CatBoost` – accuracy 0.79
- `fasttext` + `CatBoost` – accuracy 0.88
- `nn.Embedding` + `MLP` – accuracy (test) 0.78, accuracy (train) ~ 0.54

Важно отметить, что даже на обучении мы не смогли выбить качество выше 0.70, то есть модель недостаточно выразительна.

Вопрос: Как можно повысить качество?

Ответ:

Можно, например:

- Проинициализировать эмбеддинги модели какими-нибудь предобученными (например, из fasttext)
- Увеличить глубину сети
- Использовать другую архитектуру

Инициализация эмбеддингов предобученными

In [88]:
```
model_fasttext = fasttext.load_model("model_fasttext.bin")
```

In [96]:
```
class MLPClassifier(torch.nn.Module):
    def __init__(
        self,
        embedding_dim,
        hidden_dim,
        output_size,
        tokenizer,
        fasttext_model,
        dropout=0.5,
        freeze_embeddings=True # по дефолту будем считать, что эмбеддинги фа
    ):
        super().__init__()

        self.tokenizer = tokenizer
        self.hidden_dim = hidden_dim
        self.output_size = output_size
        self.embedding_dim = embedding_dim
        self.dropout = dropout

        vocab_size = self.tokenizer.get_vocab_size()
        pad_id = self.tokenizer.token_to_id('<pad>')

        self.word_embeddings = nn.Embedding(
            vocab_size,
            embedding_dim,
            padding_idx=pad_id
        )

        embedding_matrix = self._build_embedding_matrix(fasttext_model)

        self.word_embeddings.weight.data.copy_(embedding_matrix)

        if freeze_embeddings:
            self.word_embeddings.weight.requires_grad = False

        self.features = nn.Sequential(
            nn.Linear(self.embedding_dim, hidden_dim),
```

```
            nn.ReLU(),
            nn.Dropout(dropout),

            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU(),
            nn.Dropout(dropout),

            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU(),
            nn.Dropout(dropout),
        )

        self.output = torch.nn.Sequential(
          torch.nn.Linear(self.hidden_dim, self.output_size)
        )

    def _build_embedding_matrix(self, fasttext_model):
        vocab_size = self.tokenizer.get_vocab_size()

        matrix = np.zeros((vocab_size, self.embedding_dim))

        for token, idx in self.tokenizer.get_vocab().items():
            if token == "<pad>":
                continue
            matrix[idx] = fasttext_model.get_word_vector(token)

        return torch.tensor(matrix, dtype=torch.float32)

    def forward(self, tokens):
        embedding = self.word_embeddings(tokens)

        h = embedding.mean(dim=1)
        h = self.features(h)
        out = self.output(h)

        return out
```

In [90]:

```
set_global_seed(52)

device = 'cpu'

if torch.cuda.is_available():
    device = 'cuda:0'
device
```

Out[90]: 'cpu'

In [91]:

```
loss_fn = nn.CrossEntropyLoss(reduction='mean')
```

In [99]:

```
max_len    = 100
batch_size = 128
lr         = 3 * 1e-3

embedding_dim = 100
```

```
hidden_dim    = 256
output_size   = 2
```

In [93]:
```
train_dataset = TextDataset(corpus_train, y_train, tokenizer, max_len=max_le
test_dataset  = TextDataset(corpus_test , y_train, tokenizer, max_len=max_le
```

```
100%|██████████| 25000/25000 [00:53<00:00, 470.77it/s]
100%|██████████| 25000/25000 [00:48<00:00, 515.90it/s]
```

In [94]:
```
train_dataloader = DataLoader(
    train_dataset, batch_size=batch_size, shuffle=True, num_workers=0, drop_
    collate_fn=partial(collate_fn, padding_value=tokenizer.token_to_id('<pac
)
```

In [95]:
```
test_dataloader = DataLoader(
    test_dataset, batch_size=batch_size, shuffle=True, num_workers=0, drop_l
    collate_fn=partial(collate_fn, padding_value=tokenizer.token_to_id('<pac
)
```

In [100…
```
net = MLPClassifier(
    embedding_dim   = embedding_dim,
    hidden_dim      = hidden_dim,
    output_size     = output_size,
    tokenizer       = tokenizer,
    fasttext_model  = model_fasttext
)
```

In [101…
```
optimizer = torch.optim.Adam(net.parameters(), lr=lr)
```

In [102…
```
train_comet_eval(
    epoch_num        = 100,
    net              = net,
    optimizer        = optimizer,
    loss_fn          = loss_fn,
    train_dataloader = train_dataloader,
    valid_dataloader = test_dataloader,
    device           = device,
    run_name         = "MLP_classifier_with_fasttext"
)
```

```
COMET WARNING: To get all data logged automatically, import comet_ml before t
he following modules: keras, sklearn, torch, tensorflow, xgboost.
COMET WARNING: As you are running in a Jupyter environment, you will need to
 call `experiment.end()` when finished to ensure all metrics and code are log
ged before exiting.
COMET INFO: Experiment is live on comet.com https://www.comet.com/eadenisoff/
text-classifier-mmp2026/84b84a9f721f4c2aa486cf18b37302d2

COMET INFO: Couldn't find a Git repository in '/content' nor in any parent di
rectory. Set `COMET_GIT_DIRECTORY` if your Git Repository is elsewhere.
100%|██████████| 100/100 [12:42<00:00,  7.63s/it]
COMET INFO: ------------------------------------------------------------------
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COMET INFO: Comet.ml Experiment Summary
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COMET INFO:   Data:
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COMET INFO:     name                  : MLP_classifier_with_fasttext
COMET INFO:     url                   : https://www.comet.com/eadenisoff/text
-classifier-mmp2026/84b84a9f721f4c2aa486cf18b37302d2
COMET INFO:   Metrics [count] (min, max):
COMET INFO:     eval/accuracy [100]    : (0.7346554487179487, 0.7761217948717
949)
COMET INFO:     eval/loss [100]        : (0.4766505982631292, 0.5383299855085
519)
COMET INFO:     train/accuracy [19500] : (0.3828125, 0.8828125)
COMET INFO:     train/loss [19500]     : (0.33116450905799866, 0.790897607803
3447)
COMET INFO:   Others:
COMET INFO:     Name         : MLP_classifier_with_fasttext
COMET INFO:     notebook_url : https://colab.research.google.com/notebook#fil
eId=1PLmUie0LxPsu0cv-QAVB0Smqqls4aL5w
COMET INFO:   Uploads:
COMET INFO:     environment details : 1
COMET INFO:     filename            : 1
COMET INFO:     installed packages  : 1
COMET INFO:     notebook            : 2
COMET INFO:     os packages         : 1
COMET INFO:     source_code         : 1
COMET INFO:
COMET WARNING: To get all data logged automatically, import comet_ml before t
he following modules: keras, sklearn, torch, tensorflow, xgboost.
```

Видим, что даже с хорошими эмбеддингами из fasttext в случае, когда мы замораживаем их, выучить модель не удается. Попробуем теперь разморозить эмбеддинги и учить их вместе с линейными слоями.

In [103…

```
set_global_seed(52)

net = MLPClassifier(
    embedding_dim     = embedding_dim,
    hidden_dim        = hidden_dim,
    output_size       = output_size,
    tokenizer         = tokenizer,
```

```
    fasttext_model    = model_fasttext,
    freeze_embeddings = False
)
```

In [104…

```
optimizer = torch.optim.Adam(net.parameters(), lr=lr)
```

In [105…

```
train_comet_eval(
    epoch_num        = 100,
    net              = net,
    optimizer        = optimizer,
    loss_fn          = loss_fn,
    train_dataloader = train_dataloader,
    valid_dataloader = test_dataloader,
    device           = device,
    run_name         = "MLP_classifier_with_fasttext_not_freezed"
)
```

```
COMET WARNING: To get all data logged automatically, import comet_ml before t
he following modules: keras, sklearn, torch, tensorflow, xgboost.
COMET WARNING: As you are running in a Jupyter environment, you will need to
 call `experiment.end()` when finished to ensure all metrics and code are log
ged before exiting.
COMET INFO: Couldn't find a Git repository in '/content' nor in any parent di
rectory. Set `COMET_GIT_DIRECTORY` if your Git Repository is elsewhere.
COMET INFO: Experiment is live on comet.com https://www.comet.com/eadenisoff/
text-classifier-mmp2026/3d3e20004f6d4b1986e77fcb2de7b402

100%|██████████| 100/100 [15:32<00:00,  9.32s/it]
COMET INFO: ------------------------------------------------------------------
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COMET INFO: Comet.ml Experiment Summary
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COMET INFO:   Data:
COMET INFO:     display_summary_level : 1
COMET INFO:     name                  : MLP_classifier_with_fasttext_not_free
zed
COMET INFO:     url                   : https://www.comet.com/eadenisoff/text
-classifier-mmp2026/3d3e20004f6d4b1986e77fcb2de7b402
COMET INFO:   Metrics [count] (min, max):
COMET INFO:     eval/accuracy [100]    : (0.7533653846153846, 0.8103766025641
026)
COMET INFO:     eval/loss [100]        : (0.41080787212420733, 14.27486879886
725)
COMET INFO:     train/accuracy [19500] : (0.4140625, 1.0)
COMET INFO:     train/loss [19500]     : (3.2388379622716457e-06, 0.730825304
9850464)
COMET INFO:   Others:
COMET INFO:     Name         : MLP_classifier_with_fasttext_not_freezed
COMET INFO:     notebook_url : https://colab.research.google.com/notebook#fil
eId=1PLmUie0LxPsu0cv-QAVB0Smqqls4aL5w
COMET INFO:   Uploads:
COMET INFO:     environment details : 1
COMET INFO:     filename            : 1
COMET INFO:     installed packages  : 1
COMET INFO:     notebook            : 2
COMET INFO:     os packages         : 1
COMET INFO:     source_code         : 1
COMET INFO:
COMET WARNING: To get all data logged automatically, import comet_ml before t
he following modules: keras, sklearn, torch, tensorflow, xgboost.
```

Как можем заметить, хорошее начальное приближение эмбеддингов помогло модели сойтись в точку минимума на обучающей выборке, и она переобучилась. Возможно, используемая нами архитектура в целом не подходит для конкретной задачи, и нужно пробовать **что-то другое**.

Сверточная нейронная сеть (CNN)

Попробуем еще раз посмотреть, что именно мы делаем.

Общий вид архитектуры:

- Переводим токены в вектора фиксированного размера (эмбеддинги)
- Стакаем вектора в матрицу
- Извлекаем признаки из матрицы
- Подаем признаки в логистическую регрессию

Выше мы извлекали признаки не из всей матрицы, а сначала усредняли вдоль строк.

Вопрос: Зачем мы усредняли матрицу?

Вопрос: Какие способы извлечения признаков из матрицы вы знаете?

Мы можем использовать сверточные архитектуры, только с одной **оговоркой**: мы извлекаем признаки из матрицы, а не из тензора (изображения). Именно поэтому мы будем использовать `nn.Conv1d` и действовать сверткой вдоль строк (каждая строка отвечает за эмбеддинг своего токена).

Вопрос: Что позволяют учитывать свертки?

Ответы:

1. Таким образом мы получаем векторное представление всего текста, агреггирующее в себе эмбеддинги отдельных слов.
2. Можно, например, пройтись по матрице фильтром (сверткой)
3. Свертки помогают учитывать локальную структуру и взаимодействие соседних признаков

Одномерная свертка для работы с эмбеддингами

In [106…

```
embed_dim = 128

emb_layer = torch.nn.Embedding(
    tokenizer.get_vocab_size(), embed_dim,
    padding_idx=tokenizer.token_to_id('<pad>')
)

embds = emb_layer(batch['tokens'])
embds.shape
```

Out[106]: `torch.Size([4, 20, 128])`

Мы хотим подействовать на строки, поэтому нам нужно транспонировать матрицу эмбеддингов.

In [107…

```
embds = embds.transpose(1, 2)
embds.shape
```

Out[107]: torch.Size([4, 128, 20])

In [108…
```
conv_layer = nn.Conv1d(128, 32, kernel_size=3)

out = conv_layer(embds)
out.shape
```

Out[108]: torch.Size([4, 32, 18])

Вопрос: Почему последняя размерность стала 18? Какой информацию хранят новые признаки?

К сожалению, к нам приходят предложения разной длины, но мы хотим уметь работать с предложениями произвольной длины.

Вопрос: Как мы обрабатывали изображения произвольного размера? Как надо модифицировать архитектуру?

Ответы:

1. Мы проходимся сверткой с ядром 3 без паддинга $\Rightarrow$ исходный размер уменьшается на 2
2. Необходимо применить операцию пуллинга

Архитектура модели

In [109…
```
avg_pool_global = nn.AdaptiveAvgPool1d(output_size=1)

out_pool = avg_pool_global(out)
out_pool.shape
```

Out[109]: torch.Size([4, 32, 1])

На основе этой идеи построим модель.

In [110…
```
class CNNClassifier(torch.nn.Module):
    def __init__(
        self, embedding_dim, hidden_dim, output_size, tokenizer: Tokenizer,
        dropout=0.5
    ):
        super().__init__()

        self.dropout = dropout

        self.tokenizer = tokenizer
        self.hidden_dim = hidden_dim
        self.output_size = output_size
        self.embedding_dim = embedding_dim

        # Create a simple lookup table that stores embeddings of a fixed dic
```

```
        #    Use torch.nn.Embedding.
        self.word_embeddings = torch.nn.Embedding(
            self.tokenizer.get_vocab_size(), self.embedding_dim,
            padding_idx=self.tokenizer.token_to_id('<pad>')
        )

        self.features = torch.nn.Sequential(
            nn.Conv1d(self.embedding_dim, self.hidden_dim, 3), # Поменяли
            nn.ReLU(), nn.Dropout1d(self.dropout),
            nn.Conv1d(self.hidden_dim,    self.hidden_dim, 3), # Поменяли
            nn.ReLU(), nn.Dropout1d(self.dropout),
            nn.AdaptiveAvgPool1d(output_size=1),               # Поменяли
            nn.Flatten(start_dim=1)                            # Поменяли
        )
        # Create linear layer for classification
        self.output = torch.nn.Sequential(
            torch.nn.Linear(self.hidden_dim, self.output_size),
        )

    def forward(self, tokens):
        """
        :param torch.Tensor(dtype=torch.long) tokens: Batch of texts represe
        :return torch.Tensor(dtype=torch.long): Vector representation for ea
        """
        # Evaluate embeddings
        embedding = self.word_embeddings(tokens)

        # Make forward pass through CNN network
        h   = embedding.transpose(1, 2)                        # Поменяли
        h   = self.features(h)
        out = self.output(h)

        return out
```

In [111…

```
net = CNNClassifier(
    embedding_dim = 128,
    hidden_dim    = 64,
    output_size   = 2,
    tokenizer     = tokenizer
)

out = net(batch['tokens'])
out.shape
```

Out[111]: torch.Size([4, 2])

In [112…

```
print_params_count(net)
```

```
Информация о числе параметров модели: CNNClassifier
Всего параметров:                 677122
Всего обучаемых параметров:       677122
```

Обучение CNN

В случае сверточных нейронных сетей мы можем достаточно быстро переобучиться, поэтому уменьшим число параметров модели.

In [113…
```
max_len    = 100
batch_size = 128
lr         = 3 * 1e-3

embedding_dim = 32
hidden_dim    = 16
output_size   = 2
```

In [114…
```
set_global_seed(52)
```

In [115…
```
train_dataloader = DataLoader(
    train_dataset, batch_size=batch_size, shuffle=True, num_workers=0, drop_
    collate_fn=partial(collate_fn, padding_value=tokenizer.token_to_id('<pac
)
```

In [116…
```
test_dataloader = DataLoader(
    test_dataset, batch_size=batch_size, shuffle=True, num_workers=0, drop_l
    collate_fn=partial(collate_fn, padding_value=tokenizer.token_to_id('<pac
)
```

In [118…
```
set_global_seed(52)

net = CNNClassifier(
    embedding_dim = embedding_dim,
    hidden_dim    = hidden_dim,
    output_size   = output_size,
    tokenizer     = tokenizer
)
```

In [119…
```
optimizer = torch.optim.Adam(net.parameters(), lr=lr)
```

In [120…
```
train_comet_eval(
    epoch_num        = 100,
    net              = net,
    optimizer        = optimizer,
    loss_fn          = loss_fn,
    train_dataloader = train_dataloader,
    valid_dataloader = test_dataloader,
    device           = device,
    run_name         = "CNN_classifier"
)
```

```
COMET WARNING: To get all data logged automatically, import comet_ml before t
he following modules: keras, sklearn, torch, tensorflow, xgboost.
COMET WARNING: As you are running in a Jupyter environment, you will need to
 call `experiment.end()` when finished to ensure all metrics and code are log
ged before exiting.
COMET INFO: Experiment is live on comet.com https://www.comet.com/eadenisoff/
text-classifier-mmp2026/bce176769f05429dac716fa8111ede89

COMET INFO: Couldn't find a Git repository in '/content' nor in any parent di
rectory. Set `COMET_GIT_DIRECTORY` if your Git Repository is elsewhere.
100%|██████████| 100/100 [16:24<00:00,  9.85s/it]
COMET INFO: ------------------------------------------------------------------
----------------------
COMET INFO: Comet.ml Experiment Summary
COMET INFO: ------------------------------------------------------------------
----------------------
COMET INFO:   Data:
COMET INFO:     display_summary_level : 1
COMET INFO:     name                  : CNN_classifier
COMET INFO:     url                   : https://www.comet.com/eadenisoff/text
-classifier-mmp2026/bce176769f05429dac716fa8111ede89
COMET INFO:   Metrics [count] (min, max):
COMET INFO:     eval/accuracy [100]    : (0.6799278846153847, 0.7993589743589
744)
COMET INFO:     eval/loss [100]        : (0.4246353948727632, 1.2437613664529
263)
COMET INFO:     train/accuracy [19500] : (0.375, 1.0)
COMET INFO:     train/loss [19500]     : (0.02859499305486679, 0.948592007160
1868)
COMET INFO:   Others:
COMET INFO:     Name         : CNN_classifier
COMET INFO:     notebook_url : https://colab.research.google.com/notebook#fil
eId=1PLmUie0LxPsu0cv-QAVB0Smqqls4aL5w
COMET INFO:   Uploads:
COMET INFO:     environment details : 1
COMET INFO:     filename            : 1
COMET INFO:     installed packages  : 1
COMET INFO:     notebook            : 2
COMET INFO:     os packages         : 1
COMET INFO:     source_code         : 1
COMET INFO:
COMET WARNING: To get all data logged automatically, import comet_ml before t
he following modules: keras, sklearn, torch, tensorflow, xgboost.
```

При обучении нейронных сетей на маленьких выборках можно очень просто переобучить модель. Заметим, что на train мы смогли добиться почти идеального качества, значит, модель извлекает достаточно информации о выборке.

Мы бы могли повысить качество и уменьшить переобучение следующим образом:

- Добавить больше регуляризации для обучения.
- Использовать более умную токенизацию.
- Лучше предобработать текст.
- Динамически изменять лернинг рейт во время обучения.

Однако при прочих равных на нашем игрушечном эксперименте получились следующие результаты:

- `word2vec` + `CatBoost` – accuracy ~ 0.79
- `fasttext` + `CatBoost` – accuracy ~ 0.85
- `nn.Embedding` + `MLP` – accuracy (test) ~ 0.78, accuracy (train) ~ 0.54
- `fasttext` + `MLP` (freezed embeddings) – accuracy (test) ~ 0.76, accuracy (train) ~ 0.70
- `fasttext` + `MLP` (not freezed embeddings) – accuracy (test) ~ 0.77, accuracy (train) ~ 1
- `nn.Embedding` + `CNN` – accuracy (test) ~ 0.73, accuracy (train) ~ 0.96

Вопрос: Какие смысловые проблемы вы видете в сверточных сетях?

Ответ:

Свёртки выделяют локальные взаимодействия между соседними словами по отдельности, после чего вся эта информация усредняется. При этом вполне логично предположение о том, что язык нужно обрабатывать последовательно, наподобие тому, как это делают люди при чтении текста. Для этого используются архитектуры, о которых будет рассказано на следующих занятиях.

Выводы

- Использовать стандартные приёмы для получения эмбеддингов бывает полезно как в качестве **теоретического бейзлайна**, так и на **практике**.
- Можно задавать различные архитектуры: полносвязные, сверточные.
- Мы пока не научились учитывать контекст произвольной длины.